

Window Virtual Lab

Ticket 1 – No Internet Connection

Issue

User reports no internet access. Websites are not loading.

Environment

- Windows 11 VM
- Ethernet connected

Tools Used

- Command Prompt
- ipconfig
- ping

Troubleshooting Steps

- Verified IP configuration using `ipconfig`
- Tested connectivity with `ping 8.8.8.8`
- Tested DNS resolution with `ping google.com`
- Flushed DNS cache using `ipconfig /flushdns`
- Retested DNS resolution

Root Cause

Corrupted or outdated DNS cache prevented hostname resolution.

Resolution

Flushed DNS cache and restored normal DNS resolution.

Verification

Successfully pinged `google.com` and accessed websites.

Screenshots

```
Command Prompt

Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::2%4
                           10.0.2.2

C:\Users\vboxuser>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=27ms TTL=255
Reply from 8.8.8.8: bytes=32 time=23ms TTL=255
Reply from 8.8.8.8: bytes=32 time=24ms TTL=255
Reply from 8.8.8.8: bytes=32 time=23ms TTL=255

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 27ms, Average = 24ms

C:\Users\vboxuser>ping google.com

Pinging google.com [142.250.191.206] with 32 bytes of data:
Reply from 142.250.191.206: bytes=32 time=24ms TTL=255
Reply from 142.250.191.206: bytes=32 time=22ms TTL=255
Reply from 142.250.191.206: bytes=32 time=24ms TTL=255
Reply from 142.250.191.206: bytes=32 time=27ms TTL=255

Ping statistics for 142.250.191.206:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 27ms, Average = 24ms

C:\Users\vboxuser>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\vboxuser>ping google.com

Pinging google.com [142.250.191.206] with 32 bytes of data:
Reply from 142.250.191.206: bytes=32 time=25ms TTL=255
Reply from 142.250.191.206: bytes=32 time=22ms TTL=255
Reply from 142.250.191.206: bytes=32 time=23ms TTL=255
Reply from 142.250.191.206: bytes=32 time=24ms TTL=255

Ping statistics for 142.250.191.206:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 25ms, Average = 23ms

C:\Users\vboxuser>
```

Ticket 2 – Slow Computer

Issue

User reports that computer is running very slowly.

Environment

- Windows 11 VM

Tools Used

- Task Manager
- Startup Manager

Troubleshooting Steps

- Opened Task Manager and checked CPU/RAM usage
- Ended unnecessary background processes
- Disabled unnecessary startup programs
- Verified improvement in system performance

Root Cause

Too many background processes and startup programs consuming resources.

Resolution

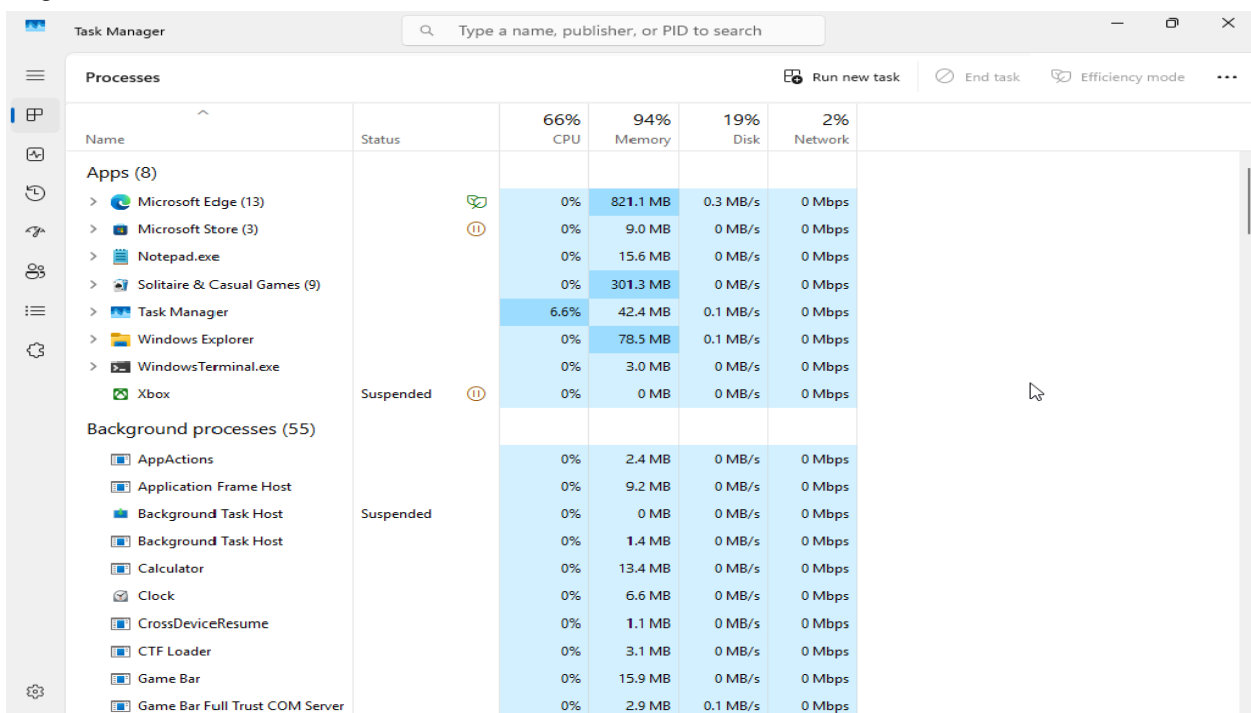
Ended unnecessary processes and disabled non-essential startup apps.

Verification

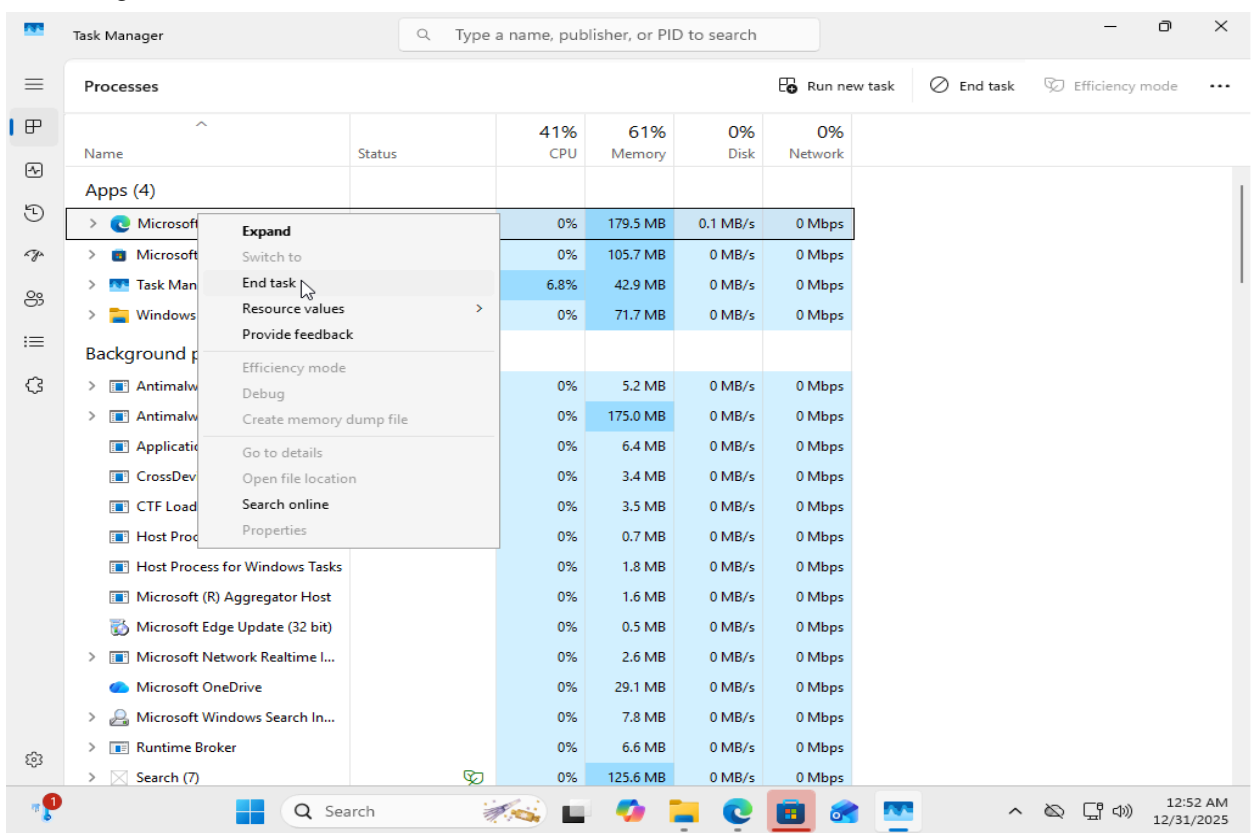
Observed faster response and reduced CPU/RAM usage.

Screenshots

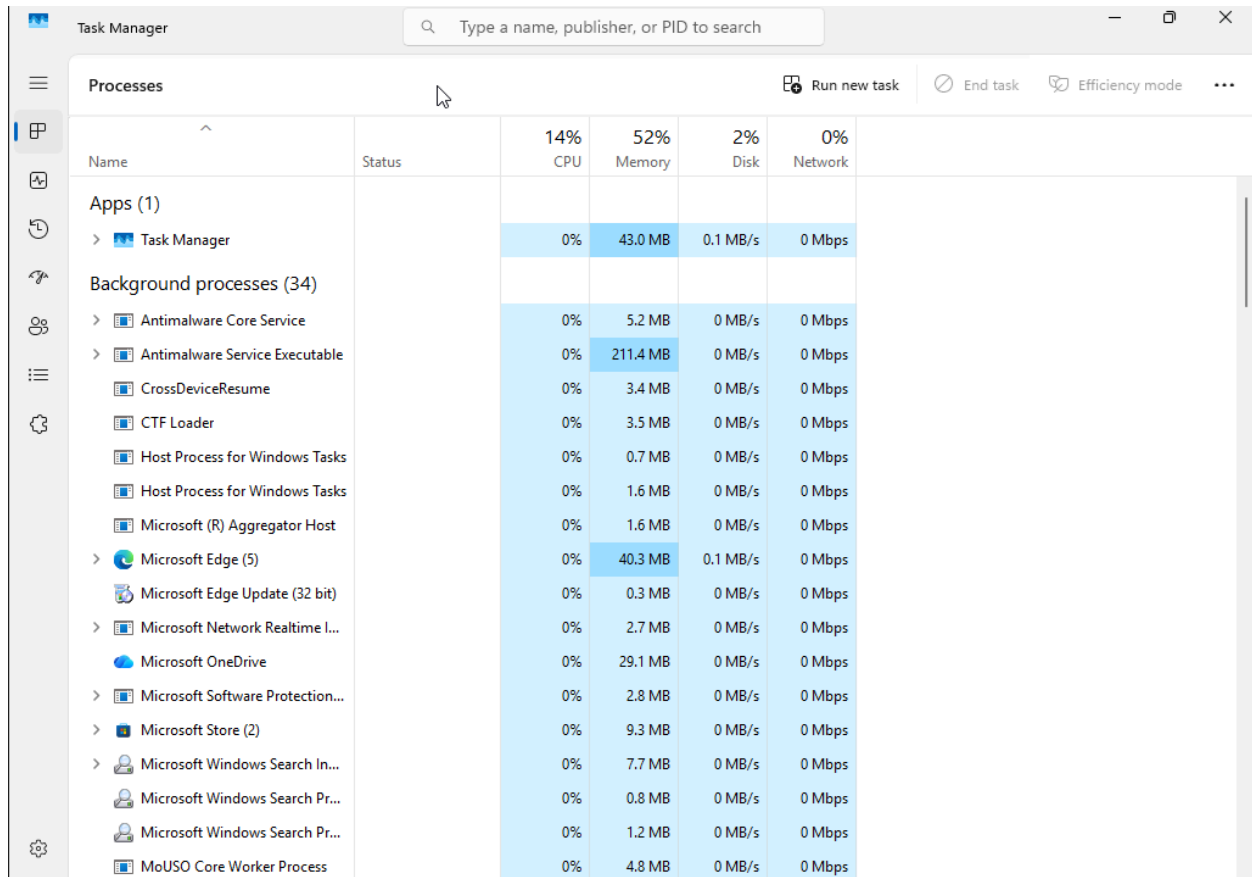
[High CPU]



[Disabling tasks]



[After Cleanup]



The screenshot shows the Windows Task Manager window with the 'Processes' tab selected. The window title is 'Task Manager' and it has a search bar at the top. The left sidebar shows icons for Task Manager, Performance, History, Startup, Services, and Settings. The main area displays a table of running processes. The table has columns for Name, Status, CPU, Memory, Disk, and Network. The processes are grouped into 'Apps (1)' and 'Background processes (34)'. The 'Task Manager' process is highlighted in blue. The 'Background processes' list includes various system services and applications like Antimalware Core Service, Microsoft Edge, and Microsoft Store.

Name	Status	14% CPU	52% Memory	2% Disk	0% Network
Apps (1)					
Task Manager		0%	43.0 MB	0.1 MB/s	0 Mbps
Background processes (34)					
Antimalware Core Service		0%	5.2 MB	0 MB/s	0 Mbps
Antimalware Service Executable		0%	211.4 MB	0 MB/s	0 Mbps
CrossDeviceResume		0%	3.4 MB	0 MB/s	0 Mbps
CTF Loader		0%	3.5 MB	0 MB/s	0 Mbps
Host Process for Windows Tasks		0%	0.7 MB	0 MB/s	0 Mbps
Host Process for Windows Tasks		0%	1.6 MB	0 MB/s	0 Mbps
Microsoft (R) Aggregator Host		0%	1.6 MB	0 MB/s	0 Mbps
Microsoft Edge (5)		0%	40.3 MB	0.1 MB/s	0 Mbps
Microsoft Edge Update (32 bit)		0%	0.3 MB	0 MB/s	0 Mbps
Microsoft Network Realtime I...		0%	2.7 MB	0 MB/s	0 Mbps
Microsoft OneDrive		0%	29.1 MB	0 MB/s	0 Mbps
Microsoft Software Protection...		0%	2.8 MB	0 MB/s	0 Mbps
Microsoft Store (2)		0%	9.3 MB	0 MB/s	0 Mbps
Microsoft Windows Search In...		0%	7.7 MB	0 MB/s	0 Mbps
Microsoft Windows Search Pr...		0%	0.8 MB	0 MB/s	0 Mbps
Microsoft Windows Search Pr...		0%	1.2 MB	0 MB/s	0 Mbps
MoUSO Core Worker Process		0%	4.8 MB	0 MB/s	0 Mbps

Ticket 3 – File Access / Permission Issue

Issue

User cannot access a shared folder. Access is denied when trying to open the folder.

Environment

- Windows 11 VM
- Local folder used for simulation

Tools Used

- Folder Properties → Security Tab

Troubleshooting Steps

- User attempted to open folder → received “Access Denied” error
- Checked folder permissions: Right-click → Properties → Security
- Added user account to allowed group / restored permissions
- Retested folder access → folder opens successfully

Root Cause

Folder permissions were incorrectly set, preventing the user from accessing it.

Resolution

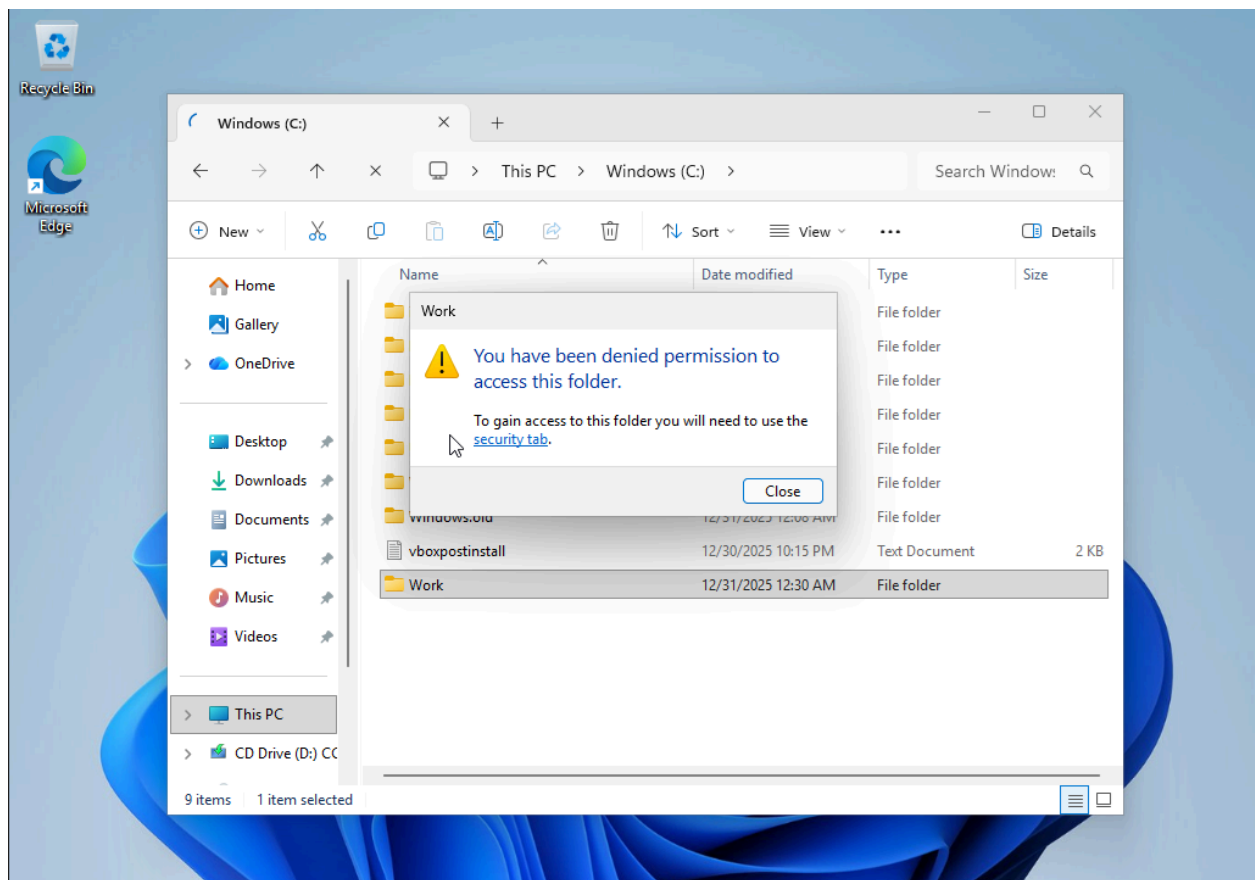
Restored proper folder permissions for the user account.

Verification

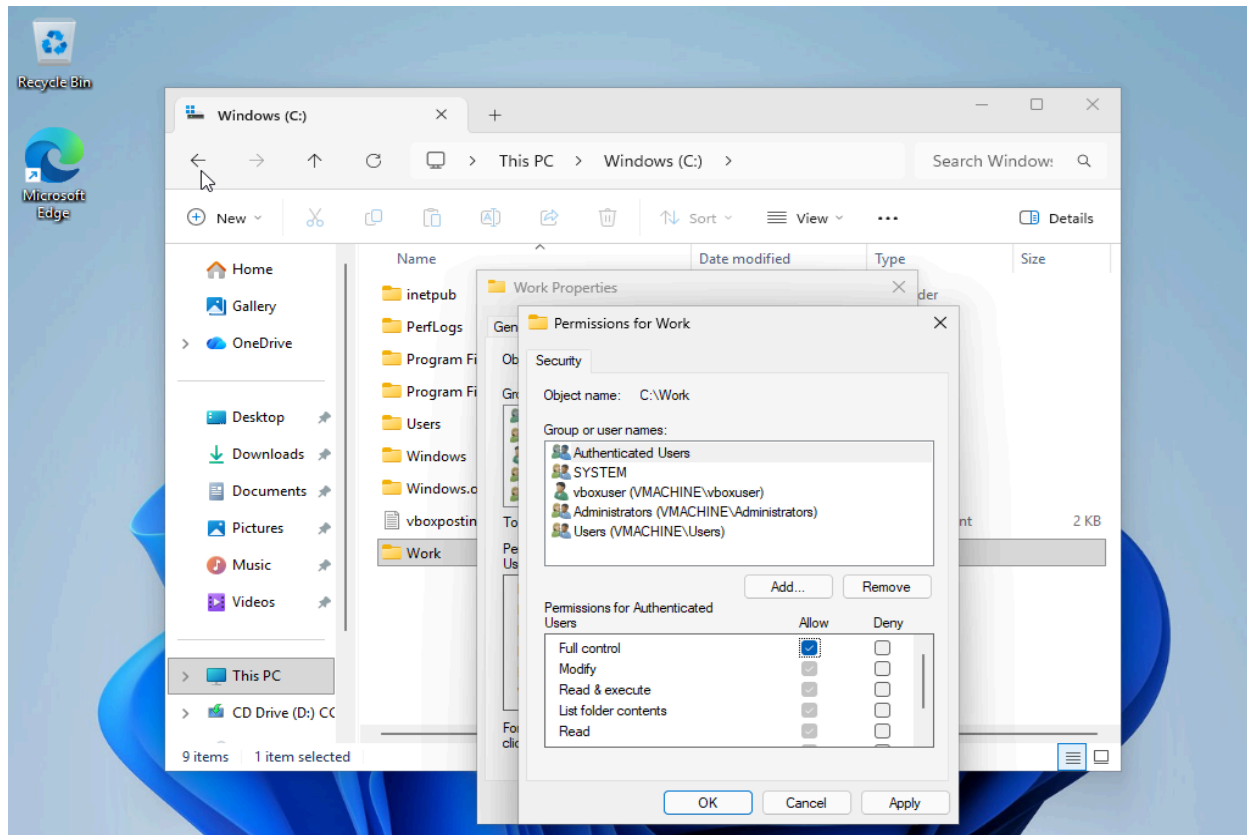
User successfully opened and accessed files in the folder.

Screenshots

[Before]



[Progress]



[After]

